22AIE205-Introduction to Python

Lab Exercise 6

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Assignment 1 — Add tags column

Code:

```
•[3]: #CH.SC.U4AIE25001
# 1.Add one more column to the existing quotes dataset to store the tags associated with each quote.
import requests
from bs4 import BeautifulSoup
import pandas as pd


url = "https://quotes.toscrape.com/"
response = requests.get(url)


soup = BeautifulSoup(response.text, "html.parser")
quote_blocks = soup.find_all("div", class_="quote")

completeData = []
for block in quote_blocks:
    text = block.find("span", class_="text").text
    author = block.find("small", class_="author").text
    tags = []
    for tag in block.find_all("a", class_="tag"):
        tags.append(tag.text)
    # print(text," - ",author," - ",tags)
    completeData.append([text,author,tags])

df = pd.DataFrame(completeData, columns=["Quote","Author","Tags"])
df
```

Output:

]:

	Quote	Author	Tags
0	“The world as we have created it is a process ...	Albert Einstein	[change, deep-thoughts, thinking, world]
1	“It is our choices, Harry, that show what we t...	J.K. Rowling	[abilities, choices]
2	“There are only two ways to live your life. On...	Albert Einstein	[inspirational, life, live, miracle, miracles]
3	“The person, be it gentleman or lady, who has ...	Jane Austen	[aliteracy, books, classic, humor]
4	“Imperfection is beauty, madness is genius and...	Marilyn Monroe	[be-yourself, inspirational]
5	“Try not to become a man of success. Rather be...	Albert Einstein	[adulthood, success, value]
6	“It is better to be hated for what you are tha...	André Gide	[life, love]
7	“I have not failed. I've just found 10,000 way...	Thomas A. Edison	[edison, failure, inspirational, paraphrased]
8	“A woman is like a tea bag; you never know how...	Eleanor Roosevelt	[misattributed-eleanor-roosevelt]
9	“A day without sunshine is like, you know, nig...	Steve Martin	[humor, obvious, simile]

Result: Tags column was successfully added to the quotes dataset.

Assignment 2 — Filter "love"

Code:

```
#2.Filter and display only the quotes where the tag is "love".
import requests
from bs4 import BeautifulSoup
import pandas as pd


url = "https://quotes.toscrape.com/"
response = requests.get(url)


soup = BeautifulSoup(response.text, "html.parser")
quote_blocks = soup.find_all("div", class_="quote")

completeData = []
for block in quote_blocks:
    text = block.find("span", class_="text").text
    author = block.find("small", class_="author").text
    tags = []
    for tag in block.find_all("a", class_="tag"):
        tags.append(tag.text)
    # print(text," - ",author," - ",tags)
    completeData.append([text,author,tags])

df = pd.DataFrame(completeData, columns=["Quote","Author","Tags"])
# rows_withlovetag = df[df["Tags"].str.contains("love")]
rows_with_love_tag = df[df["Tags"].apply(lambda tags: "love" in tags)]
rows_with_love_tag
```

Output:

	Quote	Author	Tags
6	“It is better to be hated for what you are tha...	André Gide	[life, love]

Result: First fetched all data(rows) then filtered the dataframe using lamda function(the rows with love tag).

Assignment 3 — BookMyShow

Code:

```
#Assignment 3: BookMyShow movie data
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = "https://in.bookmyshow.com/explore/home/chennai"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                  "AppleWebKit/537.36 (KHTML, like Gecko) "
                  "Chrome/114.0.0.0 Safari/537.36"
}

response = requests.get(url, headers=headers)
print("Status Code:", response.status_code)
soup = BeautifulSoup(response.text, "html.parser")
movies_data = []
movie_links = soup.find_all("a", class_="cnWwcT")

for link in movie_links:
    movie_url = link.get("href")
    if movie_url and "/movies/" in movie_url:

        # Movie Name
        name_element = link.find("div", class_="fpkEDc")
        if name_element:
            movie_name = name_element.text.strip()
        else:
            movie_name = "N/A"
        # Genre
        genre_element = link.find("div", class_="engTlI")
        if genre_element:
            genre = genre_element.text.strip()
        else:
            genre = "N/A"
        # Complete Movie URL
        if movie_url.startswith("/"):
            movie_url = "https://in.bookmyshow.com" + movie_url
        # Movie ID
        movie_id = movie_url.split("/")[-1]
        if movie_name != "N/A":
            movies_data.append([
                movie_name,
                genre,
                movie_id,
                movie_url
            ])


# Create DataFrame
df_movies = pd.DataFrame(
    movies_data,
    columns=[
        "Movie Name",
        "Genre",
        "Movie ID",
        "Movie URL"
    ]
)

# Save as CSV
df_movies.to_csv(
    "bms_chennai_movies.csv",
    index=False
)
```

```
print(df_movies.head())
print("\nTotal movies scraped:", len(df_movies))
print("\nCSV file saved as bms_chennai_movies.csv")
```

Output:

```
Status Code: 200
                Movie Name                         Genre    Movie ID  \
0                 I'm Game               Fantasy/Thriller  ET00473215
1                 Immortal       Fantasy/Horror/Thriller  ET00513702
2                Hi (2026)  Comedy/Drama/Family/Romantic  ET00509196
3  Bethlehem Kudumba Unit          Comedy/Drama/Romantic  ET00502829
4     Mirzapur: The Movie          Action/Crime/Thriller  ET00417686

                                      Movie URL
0  https://in.bookmyshow.com/movies/chennai/im-ga...
1  https://in.bookmyshow.com/movies/chennai/immor...
2  https://in.bookmyshow.com/movies/chennai/hi-20...
3  https://in.bookmyshow.com/movies/chennai/bethl...
4  https://in.bookmyshow.com/movies/chennai/mirza...

Total movies scraped: 5

CSV file saved as bms_chennai_movies.csv
```

Result: Fetched movie name, genre, movie id and movie url from bookmyshow and saved data in csv file.

bms_chennai_movies.csv

```
Movie Name,Genre,Movie ID,Movie URL
I'm Game,Fantasy/Thriller,ET00473215,https://in.bookmyshow.com/movies/chennai/im-game/ET00473215
Immortal,Fantasy/Horror/Thriller,ET00513702,https://in.bookmyshow.com/movies/chennai/immortal/ET00513702
Hi (2026),Comedy/Drama/Family/Romantic,ET00509196,https://in.bookmyshow.com/movies/chennai/hi-2026/ET00509196
Bethlehem Kudumba Unit,Comedy/Drama/Romantic,ET00502829,https://in.bookmyshow.com/movies/chennai/bethlehem-kudumba-unit/ET00502829
Mirzapur: The Movie,Action/Crime/Thriller,ET00417686,https://in.bookmyshow.com/movies/chennai/mirzapur-the-movie/ET00417686
```